

External structural disturbance is associated with increased low-acceleration residual scatter in SPARC galaxies: a residual-blind audit

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Abstract

We present a residual-blind phenomenological audit of low-acceleration SPARC rotation-curve residual scores. Coherence labels were assigned from source-backed external evidence, frozen before residual audit, and compared against a fixed rms-log endpoint. In the quality-selected sample, the primary C-minus-A median difference for the fixed low-acceleration residual score is 0.08427, with one-sided permutation $p = 0.00100$ and bootstrap 95% CI [0.04481, 0.15990]. The signal remains positive in leave-one-out, radius common-support, radius-matched, distance-matched, and distance-stratified controls. Effect-size checks show moderate A/C separation for the fixed score ($AUC = 0.771$) and for MOND/RAR-like baselines ($AUC = 0.721$ - 0.731), but not for the Newtonian baryonic baseline ($AUC = 0.506$). Residual distance/radius imbalance and observability systematics remain important caveats. A later conditional predictive-increment audit found no stable projection-specific increment in the same frozen A/C packet, and a source-frozen seven-galaxy EDGE-CALIFA morphology-proxy stress test failed its directional endpoint. These later results leave the marginal association intact but prevent interpreting it as conditional projection-specific information or independent external validation.

1 Motivation

The central question is whether external, residual-blind structural disturbance is marginally associated with a fixed SPARC residual-scatter endpoint. If the endpoint is sensitive to real astrophysical disturbance rather than only fitting noise or post-hoc grouping, externally disturbed galaxies should show larger scatter than externally regular disks.

The study avoids optimizing labels against residuals. The primary contrast is set before the audit: quality-selected C galaxies minus quality-selected A galaxies, measured by median rms-log residual with a one-sided greater alternative. B-labeled galaxies remain useful for trend and secondary checks, but the primary endpoint is the clean A/C separation. This design keeps the study closer to a residual-blind audit than to a morphology-assisted refit, and it makes a null result possible in the usual statistical sense.

2 Related Work And Context

SPARC is a widely used rotation-curve and mass-model data set for nearby disk galaxies, combining 3.6 micron photometry with HI/Halpha rotation curves. It is also central to radial-acceleration-

relation work, where observed centripetal acceleration correlates tightly with the baryonic acceleration inferred from the observed mass distribution. These results motivate low-acceleration residual scores as meaningful phenomenological objects, but they do not remove the need for disturbance and systematics checks.

The present study is closest to residual auditing and data-quality phenomenology. It is not a dark-matter halo fit, a full MOND test, or a derivation of a new gravitational law; it asks whether external structure information separates residual scatter after the score and labels are frozen. This framing is deliberately conservative, but it makes the result easier to compare with other residual scores and future resolved-HI samples. It also separates the present phenomenological audit from the broader projection program that originally motivated the fixed score.

Relevant observational issues include HI asymmetry, lopsidedness, non-circular motions, pressure support, asymmetric drift, beam smearing, and line-of-sight integration. HI asymmetry and lopsidedness are common enough in disk samples that they cannot be treated as rare pathologies. Three-dimensional tilted-ring modelling and LITTLE THINGS analyses show that such effects can materially affect recovered circular speeds in low-mass systems. The paper therefore reports distance, radius, inclination, velocity-error, Hubble-type, effect-size, and controlled-regression stress checks, treating unresolved observability as a limitation rather than as a solved nuisance.

3 Fixed Residual Model

The residual prescription is treated as a fixed model, not as a fitted galaxy-by-galaxy discovery model. It was originally motivated by a broader projection-based program, but the present paper evaluates only the frozen, reproducible residual score. A reader does not need to accept the broader theory to evaluate the audit below, because the operational question is simply whether this fixed score shows a marginal association with an external structural label. The coefficient $\alpha = 0.360$ is therefore not re-estimated here and should be read as part of the frozen score definition, not as a fitted claim of this paper.

For each SPARC radius point, the baryonic Newtonian speed scale is computed from the gas, disk, and bulge terms using fixed mass-to-light factors:

$$V_{\text{bar}}^2(R) = V_{\text{gas}}(R)|V_{\text{gas}}(R)| + \Upsilon_d V_{\text{disk}}^2(R) + \Upsilon_b V_{\text{bulge}}^2(R) \quad (1)$$

with fixed disk and bulge mass-to-light factors 0.5 and 0.7. The corresponding Newtonian acceleration scale is:

$$a_N(R) = \frac{V_{\text{bar}}^2(R)}{R} \quad (2)$$

after converting from $(\text{km/s})^2/\text{kpc}$ to SI units. The fixed projection multiplier used in this audit is:

$$F_{\text{proj}}(R) = 1 + S_\tau \alpha \ln\left(1 + \frac{a_0}{a_N(R)}\right) \quad (3)$$

The numerical constants are fixed before the audit: α is 0.360, the acceleration scale is $1.2 \times 10^{-10} \text{ m/s}^2$, and the coherence factor is fixed at its full value of one. The predicted speed is then:

$$V_{\text{model}}(R) = F_{\text{proj}}(R) V_{\text{bar}}(R) \quad (4)$$

The pointwise log residual and galaxy-level endpoint are:

$$\epsilon_i = \ln \left(\frac{V_{\text{obs},i}}{V_{\text{model},i}} \right) \quad (5)$$

$$\text{rms}_{\log} = \left(\frac{1}{N} \sum_i \epsilon_i^2 \right)^{1/2} \quad (6)$$

The formula is a fixed working prescription set in code before the residual-blind label audit. It is framed here as a reproducible residual score, not as a fully derived physical theory.

3.1 Model Motivation

The model starts from the standard SPARC baryonic decomposition rather than from observed residuals. The only additional ingredient is a fixed multiplicative correction depending on the local baryonic acceleration scale.

3.2 Rationale For The Logarithmic Ansatz

The logarithmic form of $\mathbf{F}_{\text{proj}}$ is not used as arbitrary curve-fitting convenience. It was chosen instead of a power-law boost because it remains stable in the ultra-low-acceleration regime: the correction can grow where $\mathbf{a}_N \ll \mathbf{a}_0$, but only logarithmically. It also preserves a smooth baryonic limit at high acceleration, where the logarithmic correction tends toward zero.

This choice is motivated by the broader projection picture behind the fixed score: scale-free geometric projection effects suggest dependence on ratios of acceleration scales rather than on an added absolute acceleration. A logarithm of $1 + \mathbf{a}_0/\mathbf{a}_N$ is a minimal way to encode such a ratio while preserving a smooth baryonic limit at high acceleration. In this paper that motivation defines a fixed dynamical score; it is not used as a complete derivation of the underlying theory. The audit therefore tests an operational consequence of the score, not the full theoretical program.

The acceleration scale is fixed at the familiar low-acceleration value used in galaxy-dynamics phenomenology, and $\alpha = 0.360$ is treated as a fixed model constant. Neither is fitted in the residual-blind label audit. The coherence factor is set to one, so no galaxy-specific coherence parameter is tuned to improve individual fits. The coefficient and ansatz are therefore part of the pre-frozen score; their deeper theoretical motivation is left to a companion model note.

The test is intentionally narrow: if the frozen endpoint is sensitive to structural disturbance, externally disturbed or coherence-poor galaxies should tend to have larger residual scatter than externally regular disks.

The analysis has four fixed steps:

1. Compute SPARC residual summaries under the fixed low-acceleration prescription.
2. Assign external coherence labels without inspecting those residual summaries.
3. Freeze the quality selection, labels, endpoint, and primary contrast.
4. Test whether quality-selected C galaxies have higher rms-log residual scatter than quality-selected A galaxies.

Fixed logarithmic projection formula: two contrasting cases

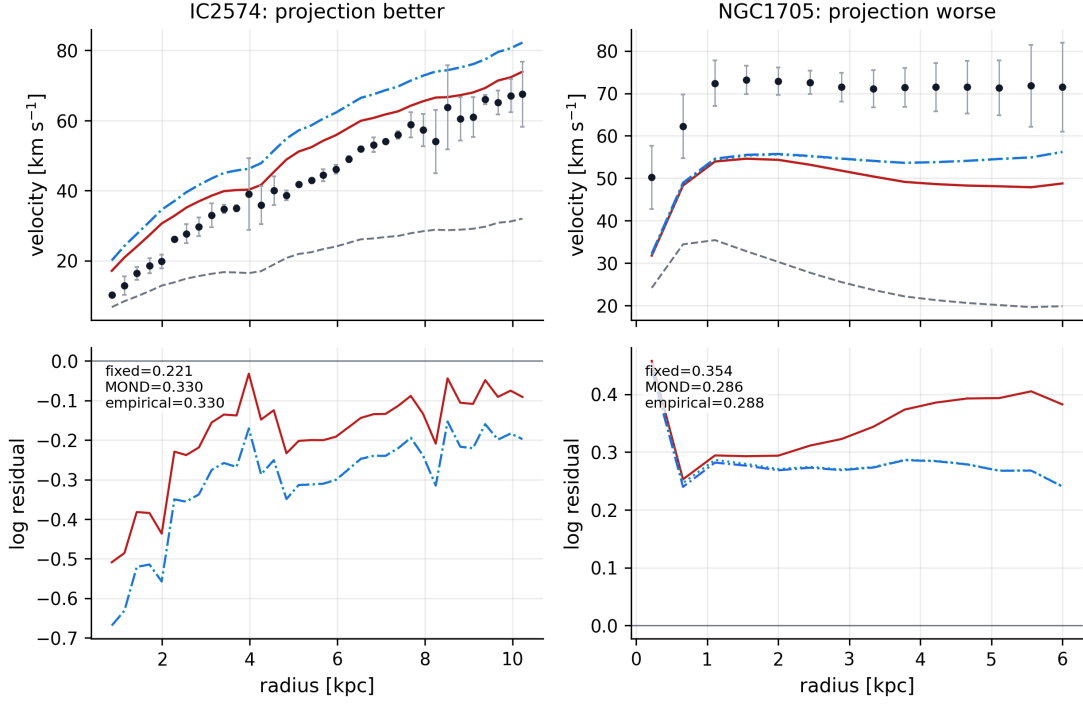


Figure 1: Fixed logarithmic projection formula compared with MOND/RAR-family baselines on two contrasting SPARC rotation curves. IC2574 illustrates a case where the fixed projection score has lower rms-log residual than the MOND/RAR-like baselines; NGC1705 illustrates the opposite. These examples are selected only to show model behavior around the logarithmic ansatz introduced above. They are not used as endpoints, tuning targets, or evidence for uniqueness over MOND/RAR-family scores. No separate numerical RMOND model is evaluated in this packet.

Near-tie behavior for the fixed logarithmic projection formula

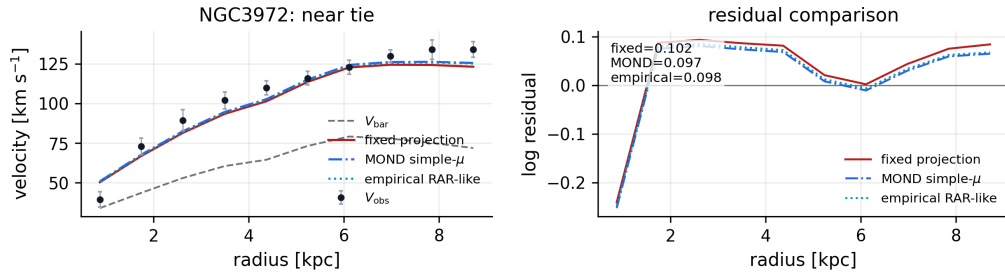


Figure 2: Near-tie behavior for NGC3972. In this case the fixed projection, MOND simple- μ , and empirical RAR-like curves give very similar residual scales, illustrating why the paper treats the fixed logarithmic prescription as an operational score rather than as a unique model-selection result.

4 Data And Endpoint

The residual source is the SPARC residual summary for the fixed low-acceleration score. The endpoint is rms-log scatter after the study quality gate, which requires sufficient rotation-curve support and diagnostic quality. The quality-selected sample contains 73 galaxies: 17 A, 28 B, and 28 C.

The primary statistic is the median difference:

$$\Delta_{AC} = \text{median}(\text{rms}_{\log} \mid C, \text{selected}) - \text{median}(\text{rms}_{\log} \mid A, \text{selected}) \quad (7)$$

Inference uses count-preserving label permutation for the one-sided p-value and bootstrap resampling for the confidence interval. This keeps the test aligned with the frozen A/C class sizes and avoids model-fitting degrees of freedom. The statistic is intentionally simple, because the sample is modest and the main risk is not underfitting but over-interpreting a flexible analysis.

External-label rotation-curve context

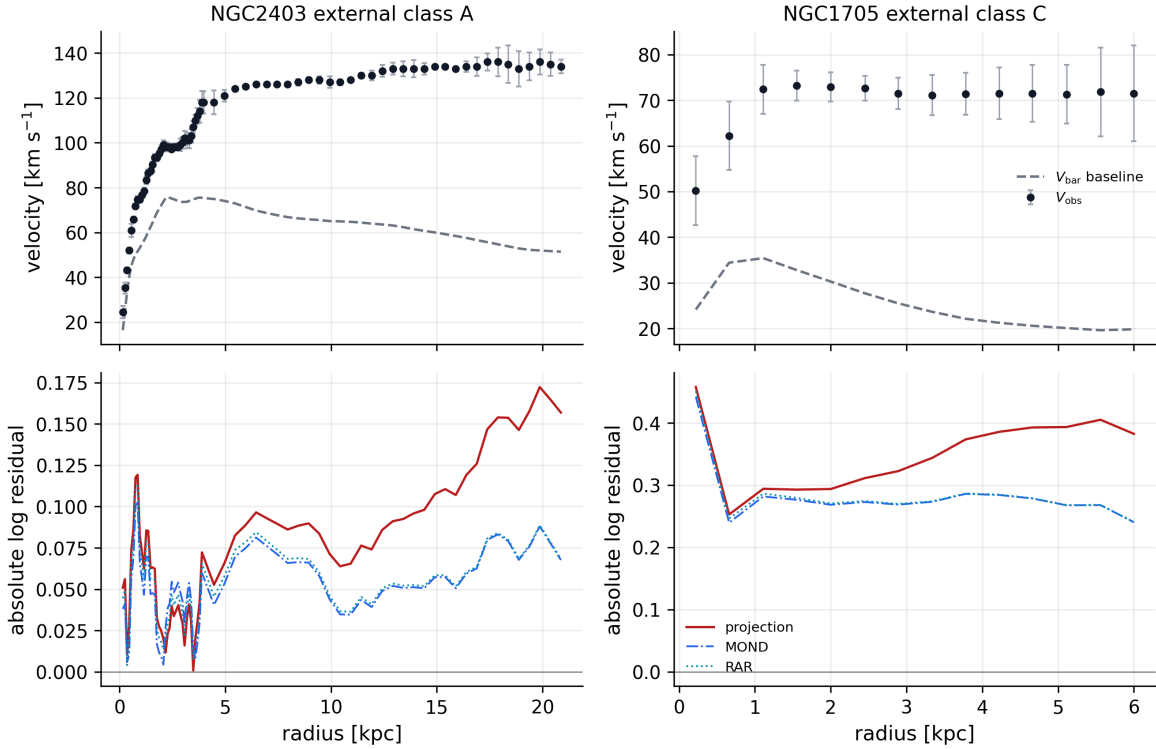


Figure 3: External-label rotation-curve context for NGC2403, a regular class-A disk, and NGC1705, a disturbed class-C dwarf. The upper panels show observed rotation curves against the baryonic baseline; the lower panels show the per-radius absolute residual profiles that feed the score families. This visualization is not a primary endpoint and is not used for labeling, threshold selection, or model tuning.

5 Labeling Protocol And External Evidence

Labels encode external structural state rather than residual behavior. Class A denotes externally regular/coherent disks, class C denotes externally disturbed or coherence-poor systems, and class B denotes uncertain or intermediate cases. The A/C evidence rows were expanded to reduce dependence on uncertain labels and to make the distance imbalance testable.

This discipline is central: the result is not that an analyst can separate high- and low-residual galaxies after inspecting residuals, but that a residual-blind external label is marginally associated with the frozen endpoint. The packet records this audit layer in `labeling_protocol.md`, which defines the source rules and A/B/C decision logic, and in `external_evidence_table.csv`, which gives row-level evidence summaries, source links, reviewer/date fields, and residual-blind flags.

The protocol is intentionally conservative. A requires direct regular-disk evidence such as regular HI/H α kinematics, low asymmetry, or a well-defined symmetric rotation curve. C requires direct disturbance evidence such as disturbed HI morphology, lopsided velocity fields, tidal or interaction signatures, or warp evidence tied to gas morphology or kinematics. Environment-only, bar-only, or rotation-curve-quality-only information is not enough for an A/C call; ambiguous rows default to B. This makes some classifications less aggressive, but it reduces the risk that the final contrast is created by weak or indirect evidence.

Figure 1 shows the quality-selected residual distribution by external class. Labels are external; plotted residuals enter only after the labels are frozen.

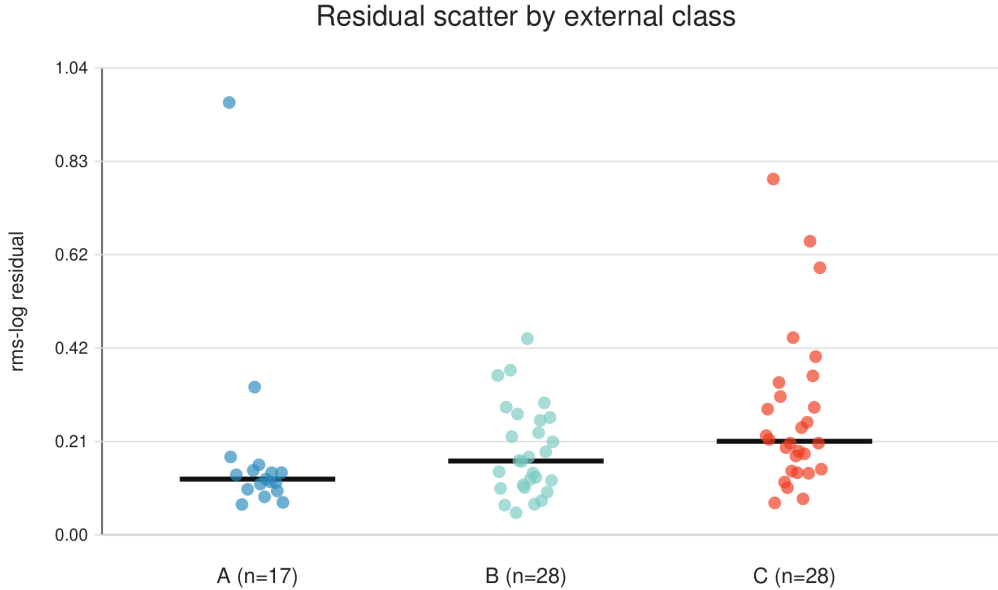


Figure 4: Residual scatter by external class.

6 Primary Result

The primary result is positive:

Externally disturbed/coherence-poor galaxies have larger fixed-score residual scatter than externally regular galaxies. The bootstrap interval is fully positive and the one-sided permutation

comparison	n A	n C	median A	median C	C-A diff	p	CI low	CI high
quality-selected C-minus-A	17	28	0.12436	0.20862	0.08427	0.00100	0.04481	0.15990

p-value is 0.00100.

The effect is not only a p-value result. A descriptive effect-size appendix gives $AUC = 0.771$ for the fixed low-acceleration residual score: a randomly selected C galaxy has larger fixed-score scatter than a randomly selected A galaxy in about 77% of A/C pairs. The robust median-scaled effect is 0.938. These quantities are descriptive effect sizes for an observational audit, not estimates of a universal physical constant.

Secondary quality-selected checks support the same ordering. The C-minus-nonC contrast is positive ($p = 0.0173$), and the ordered A-to-B-to-C trend is positive ($\rho = 0.3620$, $p = 0.00110$). The A/C leave-one-out check produces no sign flips, so the primary direction is not set by a single influential galaxy.

Figure 2 summarizes the primary result and main controls on a common effect scale.

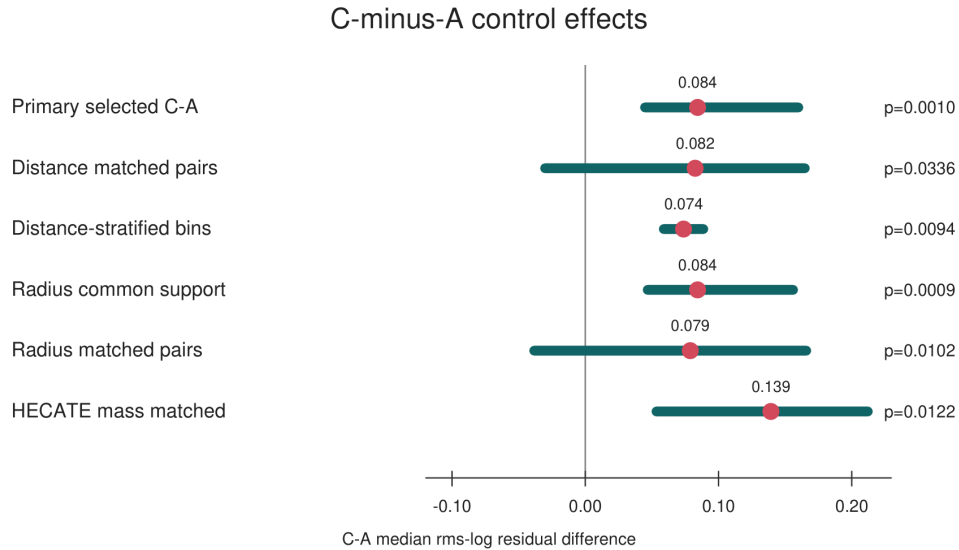


Figure 5: C-minus-A control effects.

7 Distance And Scale Controls

The main potential confound is that C galaxies are, on average, closer and smaller in radius than A galaxies. Controls reduce this concern but do not remove it. Raw SPARC distance imbalance remains visible: A median distance is 13.8 Mpc, C median distance is 7.83 Mpc, and the two-sided imbalance p-value is 0.02490. This imbalance is scientifically important because the same observability differences that make disturbance easier to identify may also change the measured rotation-curve residual structure.

The important change is that matched and stratified controls no longer collapse the signal:

Every distance bin has a positive C-minus-A median difference:

control	effect	p
greedy SPARC distance-matched pairs	0.08247	0.03360
optimal ordered SPARC distance-matched pairs	0.08247	0.03610
strict ≤ 2 Mpc distance caliper	0.11949	0.08979
distance-stratified within-bin test	0.07374	0.00940

SPARC distance bin	n A	n C	C-A diff
0-5 Mpc	3	10	0.07409
5-10 Mpc	4	10	0.06147
10-20 Mpc	7	6	0.08512
>20 Mpc	3	2	0.06361

Figure 3 shows the same within-bin result visually.

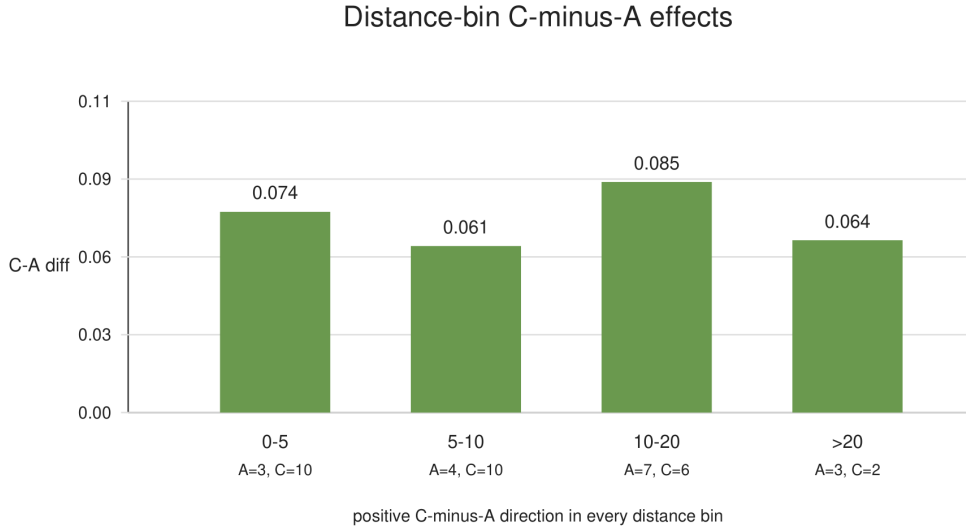


Figure 6: Distance-bin effects.

Radius controls also preserve the predicted direction. The max-radius imbalance remains visible ($p = 0.00190$), but common-support and greedy radius-matched tests are positive ($p = 0.00090$ and $p = 0.01020$); the strict ≤ 2 kpc caliper is positive but underpowered ($p = 0.10969$). Thus the radius checks support the same direction, while also showing that very strict matching quickly reduces sample support.

HECATE stellar-mass matching is directionally supportive where coverage exists ($p = 0.01220$ and $p = 0.01440$), but incomplete dwarf-regime coverage makes it supplementary.

The selection/observability appendix records distance, distance uncertainty, radial coverage, point count, velocity-error, inclination, Hubble-type, WHISP-family coverage, and partial HECATE mass balance. A robust regression stress test is positive in class-only and distance-only specifications, weakens after distance and radial coverage are included together, and remains positive but not bootstrap-exclusive of zero in the full observability specification. This is an important constraint on the claim: the signal is directionally stable across several controls, but it is not yet an observability-proof physical detection.

Threshold-sensitivity and weighted-residual figures are useful supplementary diagnostics, but they repeat the same class-ordering message rather than testing the main confound.

8 Alternative Baseline Scores

The same residual-blind A/C labels were compared against three alternative baseline scores from the same SPARC rotmod rows and fixed mass-to-light factors. These are not replacement endpoints; they test whether class separation is specific to the frozen low-acceleration score or reflects broader disturbance sensitivity.

score	n A	n C	median A	median C	C-A diff	p	CI low	CI high
fixed score	17	28	0.12436	0.20862	0.08427	0.00210	0.04440	0.15792
Newtonian baryonic	17	28	0.57056	0.61615	0.04559	0.39546	-0.20123	0.31376
MOND	17	28	0.11366	0.23238	0.11872	0.00190	0.00305	0.17644
simple-mu empirical RAR	17	28	0.11360	0.22442	0.11082	0.00210	0.00583	0.17395

The Newtonian baryonic baseline is not a significant separator. MOND-like and empirical RAR baselines show positive A/C separation, so the result should not be described as uniqueness evidence for the projection formula. The effect-size comparison sharpens this: fixed score AUC = 0.771, MOND simple-mu AUC = 0.721, empirical RAR AUC = 0.731, and Newtonian baryonic AUC = 0.506. Thus the A/C separation appears mainly in low-acceleration residual scores rather than in every residual construction. This is a useful physical constraint: the signal is not simply that disturbed galaxies look worse under any arbitrary score.

The same point can be summarized as a compact effect-size sanity check:

score	AUC	robust median d	interpretation
fixed low-acceleration score	0.771	0.938	moderate A/C separation
Newtonian baryonic	0.506	0.136	near non-informative
MOND simple-mu	0.721	1.234	compatible
empirical RAR	0.731	1.224	low-acceleration separation compatible
			low-acceleration separation

9 Interpretation

An externally assigned, residual-blind structural label separates the fixed endpoint in the predicted direction. The signal does not depend on B labels, survives leave-one-out checks, and remains positive under distance/radius/mass controls.

The conservative interpretation is a reproducible residual-disturbance association in this audited SPARC sample: externally disturbed galaxies show larger fixed-score residual scatter than externally regular disks.

The physical interpretation is not that one projection-motivated score is uniquely selected. Larger C-class scatter may mark non-equilibrium structure, non-circular motions, lopsided gas kinematics, warps, or interaction-driven asymmetries. Such systems are harder for any smooth

axisymmetric rotation-curve prescription to describe. The C>A residual scatter is therefore best read as disturbance-sensitive residual phenomenology, not model-specific proof.

Two downstream audits narrow this interpretation without changing any Paper 1 statistic. First, a retrospective Paper 3 fixed-model conditional predictive-increment audit used the same frozen 45-galaxy A/C sample, repeated stratified five-fold cross-fitting with 20 repetitions, endpoint-residual-free observability/baryonic covariates, and residual structure common to MOND/RAR-like baselines. The unique projection contrast supplied no stable held-out proper-score increment under either declared baseline and any of the three declared regularization values. At the primary O7B7, C=0.3 setting, the AUC change was -0.01890756, the log-loss improvement was -0.03994321, and the Brier-score improvement was -0.01188683; negative improvement means worse performance. This is an operational conditional predictive-increment result, not a literal conditional-mutual-information estimate and not a theorem that all additional rotation information is absent.

Second, a source-frozen seven-galaxy EDGE-CALIFA rotation-morphology proxy endpoint returned mean D=-0.058679293503004035, exact one-sided p=0.6015625, median D=+0.11020194395988532, and 4/7 positive galaxy-level values. It therefore failed its directional prevalidation gate. This is an external morphology-proxy stress test, not an independent Tau-specific test and not a direct replication of the Paper 1 A/C endpoint.

The subsequent public-data ladder also remains below claim-raising status. A prospective LIT-TLE THINGS replay transfers one frozen scale-tail family to 14 galaxies and 313 points, but gives only a 0.101 km/s mean RMSE advantage over TPG/v6, is 0.649 km/s worse than MOND, and cannot test multi-family attribution. In PHANGS, a three-galaxy morphology-orthogonal low-order CO-Halpha statistic is consistent with zero (p=0.8082); a separate two-galaxy m=2 statistic is numerically non-null but its m=1 control is also non-null and its source-label integrity audit fails. A higher-dimensional four-galaxy PHANGS confirmatory packet stops at its predeclared azimuth-sector occupancy gate and releases no score. These results constrain different proxy endpoints rather than directly repeating the Paper 1 A/C statistic.

The claim boundary after these audits is consequently an unconditional or marginal within-SPARC association. Paper 1 does not establish conditional projection-specific information, an independently replicated morphology law, or a physical Tau mechanism. The additional public-data routes strengthen the falsification record, not the physical claim.

These failures must also be read at the resolution actually tested. Each audit freezes a finite morphology representation Φ_k , not a source-complete parent body. A terminal law can descend through Φ_k only if two body states with the same Φ_k value always have the same terminal image. Failure therefore constrains the declared representation and descent, not every possible richer morphology. This distinction is not a license for post-hoc rescue: once terminal vector y is known, appending y to a source span makes its projected residual zero by construction. Any richer representation must be built from source-only information, frozen, and tested on a new untouched packet.

The later Tau Core full-4D descent does not alter any statistic reported here. It only widens the possible upstream interpretation of a terminal residual: body structure, lapse, shift, spatial geometry, source-observer photon transfer, and ordinary astrophysical or observational effects may contribute. Because no nonzero radial Tau coframe law has been derived or frozen, this audit does not identify any of those origins and must not be retrospectively rescored with a fitted 4D correction. The exact later scoring compiler strengthens this prohibition: a mixed parent Hessian is not itself a signal, and the scoreable source object would be

$$E_K = (K_{HH} - K_{std}) - CK_V^{-1}C^\dagger. \quad (8)$$

No such source-frozen standard excess or joint terminal prediction is present here.

10 Limitations

Residual distance and radius imbalance remains the main methodological limitation. Controls mitigate the risk, but raw A/C distance and radius distributions are still not exchangeable. Closer galaxies may reveal disturbances and rotation-curve substructure more readily. Strict caliper tests are positive but have limited support, and HECATE mass controls are incomplete. This remains an important caveat even though the matched and stratified controls preserve the sign of the effect.

The packet lacks homogeneous beam-size or physical HI-resolution measurements. Distance, radius, point count, and velocity-error controls are therefore observability proxies rather than a direct beam-smearing correction. The controlled-regression appendix supports the direction of the effect in simpler robust specifications but shows sensitivity to joint observability controls. This matters physically because closer or better-resolved galaxies may reveal disturbance and rotation-curve substructure more readily than more distant systems.

Inclination and kinematic systematics are not exhausted. The quality gate removes the most extreme inclination failures, but edge-on integration, low-inclination deprojection uncertainty, asymmetric drift, beam smearing, pressure support, and non-circular motions can still affect galaxy-level scatter. Modern HI analyses often use three-dimensional tilted-ring or cube-level modelling to control these effects because beam smearing and pressure-supported gas can change the recovered circular speed, especially in low-mass systems. The inclination/systematics appendix shows $C > A$ in populated inclination bins, but with small counts.

The result should therefore be read as a residual-disturbance audit, not as unique model selection. A central caveat is that C likely contains more non-equilibrium or non-circular systems, for which any smooth axisymmetric score may perform worse. This is not merely a nuisance detail; it is one plausible astrophysical origin of the observed excess scatter. Baseline comparisons sharpen the interpretation: Newtonian baryonic residuals do not reproduce the sharp A/C separation, while MOND-like and empirical RAR scores show compatible positive separation. The later conditional projection-contrast audit and the small EDGE-CALIFA proxy stress test are both negative; independent matched-tracer validation therefore remains open.

11 Known Limitations And Phase II Validation Plan

The remaining weaknesses are mainly data-side limitations. The primary A/C sample is modest ($n_A = 17$, $n_C = 28$), observability degeneracy is not fully removed, and the result is still internal to SPARC.

These limitations define Phase II. A follow-up should freeze the same endpoint, labeling rule, and effect-size summaries before applying them to independent resolved-HI samples. The target samples and observability fields should be specified before the external test:

The goal is not to tune the score, but to ask whether residual-disturbance separation reproduces under independent observability conditions.

The seven-galaxy EDGE-CALIFA morphology-proxy stress test does not satisfy this Phase II requirement: it used a different proxy endpoint, failed its directional gate, and was not an independent Tau-specific test. A claim-raising replication still requires a new source-frozen matched-tracer cohort with the nuisance and comparator structure fixed before endpoint inspection.

If the separation reproduces after endpoint and labeling rules are frozen, the result would move from a SPARC audit signal to broader galaxy-dynamics phenomenology. If not, it should be treated

sample family	main use in Phase II	minimum added metadata
WHISP	HI asymmetry and morphology stress test	beam size, distance, HI asymmetry flag
LITTLE THINGS	dwarf-galaxy and pressure-support stress test	beam size, inclination, asymmetric-drift treatment
THINGS	high-quality nearby disk comparison	velocity-field quality, non-circular motion flags
LVHIS	local-volume resolution/selection check	beam-to-disk scale, distance, inclination

as SPARC-specific or observability-driven.

12 Figures And Tables

The minimum main-text figure set is:

figure	file	role
Figure 1	<code>figures/quality_pass_rms_distribution.svg</code>	quality-selected class-separated primary endpoint
Figure 2	<code>figures/control_forest_plot.svg</code>	primary and control effects on one scale
Figure 3	<code>figures/distance_stratified_effects.svg</code>	distance-bin directionality check

The weighted-residual distribution, threshold-sensitivity heatmap, process diagram, and single-effect summary graphic are supplementary diagnostics. For a short submission, the main-text table focus should be the primary endpoint, baseline/effect-size comparison, and compact control summary; row-level evidence, regression tables, and systematics counts can remain supplementary.

The key supplementary tables are `external_evidence_table.csv` for row-level label evidence, `baseline_score_comparisons.md` for alternative-score checks, `selection_observability_appendix.md` and `controlled_regression_appendix.md` for observability controls, and `effect_size_appendix.md` plus `inclination_systematics_appendix.md` for effect-size and kinematic-systematics summaries.

13 Reproducibility Bundle

The full reproducibility package, including the frozen labels, external evidence table, baseline-score comparisons, control summaries, figures, and regeneration scripts, is archived at [doi:10.5281/zenodo.20285859](https://doi.org/10.5281/zenodo.20285859). The analysis can be regenerated with the commands listed in Section 13.

The reproducibility bundle contains generated tables, control summaries, matched-pair rows, figures, manifest, and status notes under:

```
studies/sparc_residual_coherence_test_v01/paper_packet_v06_distance_balanced
```

Key audit files include `final_tables.md`, `external_evidence_table.csv`, `baseline_score_comparisons.md`, `selection_observability_appendix.md`, `controlled_regression_appendix.md`, `effect_size_appendix.md`, and `inclination_systematics_appendix.md`.

The regeneration commands are:

```
python studies/sparc_residual_coherence_test_v01/download_sparc_data.py
python studies/sparc_residual_coherence_test_v01/make_labeling_protocol_and_baselines.py
python studies/sparc_residual_coherence_test_v01/make_selection_and_regression_appendix.py
python studies/sparc_residual_coherence_test_v01/make_effect_size_and_systematics_appendix.py
python studies/sparc_residual_coherence_test_v01/make_manuscript_pdf.py
PYTHONPATH=src python -m pytest -q
```

Raw SPARC rotmod files are not redistributed. The download script fetches the public SPARC Zenodo archive and places inputs at the paths expected by the regeneration scripts.

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14 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer–source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless

derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer Q_{OS} .

Ownership and use. The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

Consequence for this paper. For galactic inference, these observer constructions do not derive a rotation-curve correction or identify a measured residual as a parent effect. Existing endpoint freezes and scores are unchanged.

15 Source identification and recovery

The current observer audit separates a conditional forward construction from physical source selection. A frozen body, occupied observer incidence, preparation, complete instrument and stable resolution can determine records without selecting those inputs from Nature. Equal effects do not fix postmeasurement states, and symmetry within a chosen interaction family does not exclude other symmetric couplings. For galactic inference, these observer constructions do not derive a rotation-curve correction or identify a measured residual as a parent effect. Existing endpoint freezes and scores are unchanged.

Dependency and claim ownership. Foundation VI and the body–readout technical companion own the finite instrument/access proof; Foundation IV and TD-0 retain physical source selection; Foundation V, the temporal synthesis and seed–body companion own the gradient/inertial distinction. This paper inherits only the stated specialization. Canonical assumptions, countermodels and calibration audits are in https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_rqm_instrument_to_record_bridge_001.md. No physical seed, Nature occupation or Tau-exclusive observation is established by this update. Existing empirical endpoint definitions and scores are unchanged.

16 Conditional source preparation and time-resolved access

The common-operator scalar–Dirac completion now supports an explicit conditional chain from interaction vertices to pair transfer and joint readout. Nonzero vertices do not select occupation. Relational coherence can be generated from an occupied number-diagonal pair even while both

local mean fields vanish. In a declared resonant rotating-wave and Markov loss model, finite-window emission retains this coherence information, whereas an all-time count discards it. This is standard-compatible quantum recovery, not a Tau-specific observational residual. For galactic inference, these observer constructions do not derive a rotation-curve correction or identify a measured residual as a parent effect. Existing endpoint freezes and scores are unchanged.

Dependency ownership and limits. The P4 common-source assembly owns the scalar/Yukawa benchmark and finite preparation calculations; Foundation VI and the body-readout companion inherit the joint-access mathematics. Foundation IV and TD-0 retain physical source selection, and Foundation V retains clock descent. The parent remains atemporal: the evolution above is a conditional downstream 4D model, not parent memory or a derived morphogenetic time. Physical initial occupation, a gauge-complete realization, detector amplification, absolute calibration and stable observer-relative Q_{OS} remain unselected. No empirical endpoint or score is changed. The canonical derivation and reproducible numerical sources are linked at https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_core_p4_nonabelian_particle_common_source_assembly_001.md.

17 Conclusion

The distance- and scale-controlled analysis finds a positive association between external structural-disturbance labels and fixed-score residual scatter. Externally disturbed or coherence-poor galaxies show larger scatter than externally regular disks, and the direction persists under matched and stratified controls.

The result indicates a reproducible marginal residual-disturbance association in the audited SPARC sample. A natural interpretation is that C-class systems contain more non-equilibrium or non-circular motion, increasing scatter for smooth axisymmetric models. Because distance/radius imbalance remains, MOND/RAR-like baselines show compatible disturbance sensitivity, the later conditional projection-contrast audit is negative, and the seven-galaxy EDGE-CALIFA morphology-proxy stress test failed, this is not standalone proof of the projection model. The supported statement is narrower: the frozen residual score is unconditionally associated with residual-blind external labels in this SPARC sample; conditional projection-specific information and independent external replication remain unestablished.